

PARTS LIST*Semiconductors:*

- IC1 - 7805, 5V regulator
- IC2 - AT89C51 microcontroller
- T1 - BC547 npn transistor
- BR1 - 1A bridge rectifier
- LCD1 - 16×2 LCD

Resistors (all 1/4-watt, ±5% carbon):

- R1-R5 - 10-kilo-ohm
- R6 - 1-kilo-ohm
- R7 - 100-ohm
- VR1 - 10-kilo-ohm preset

Capacitors:

- C1 - 1000µF, 25V electrolytic
- C2-C3 - 0.33µF ceramic
- C4-C5 - 22pF ceramic
- C6 - 10µF, 16V electrolytic

Miscellaneous:

- S1, S2, S3 - Tactile switch
- S4 - On/off toggle switch
- X_{osc}1 - 12MHz crystal oscillator
- PZ1 - Piezo buzzer
- X1 - 230V AC primary to 9V, 500mA secondary transformer

the circuit on the PCB to save time and minimise assembly errors. Carefully assemble the components and double-check for any overlooked errors.

Initially, the user has to set the timer in off mode using switch S4. To start the timer, switch on S4 and use switch



Fig. 4: Timer on message on the LCD when switch S4 is closed

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

S3 to set the countdown timer (CNT). You can set the timer from one to 59. Fig. 4 shows the display settings with real-time clock as 07 hours, 10 minutes and 38 seconds in AM and the countdown timer (CNT) is set to one. When the timer is set, you can switch on the

electrical appliance such as a motor pump.

The real-time seconds keep on incrementing from 38 till these reach zero. As soon as the seconds reach zero (after 59), the minutes will increase to 11 and, at the same time, the countdown timer will be zero. When the countdown timer reaches zero, a buzzer

alerts the user to switch-off the electrical appliance. The buzzer continues to beep till switch S4 is switched-off. For troubleshooting, ensure that the voltages at test points are as per test points table.

This timer can also be used as an appliance timer or kitchen timer. You can even replace the buzzer with a relay for switching on/off the appliance through it. ●

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